Database Technologies – Assignment 3

Note : To solve below queries use “hr” database

1. Write a query to get unique department ID from employee table.

```
mysql> SELECT DISTINCT DEPARTMENT_ID FROM employees;
+---------------+
| DEPARTMENT_ID |
+---------------+
|          NULL |
|            10 |
|            20 |
|            30 |
|            40 |
|            50 |
|            60 |
|            70 |
|            80 |
|            90 |
|           100 |
|           110 |
+---------------+
12 rows in set (0.001 sec)
```

2. Write a query to get all the employee’s details from the employees table displaying the first names in descending order.

```
mysql> SELECT FIRST_NAME FROM employees ORDER BY FIRST_NAME DESC;
+-------------+
| FIRST_NAME  |
+-------------+
| Winston     |
| William     |
| William     |
| Vance       |
| Valli       |
| Trenna      |
| TJ          |
| Timothy     |
```

```
| Anthony      |
| Amit         |
| Alyssa       |
| Allan        |
| Alexis       |
| Alexander    |
| Alexander    |
| Alberto      |
| Alana        |
| Adam         |
+--------------+
107 rows in set (0.001 sec)
```

3. Write a query to get the employee ID, names (first_name and last_name), salary in ascending order of salary.

```
mysql> SELECT EMPLOYEE_ID, CONCAT(FIRST_NAME, ' ',LAST_NAME) AS FULL_NAME, SALARY FROM employees ORDER BY SALARY;
+-------------+------------------+----------+
| EMPLOYEE_ID | FULL_NAME        | SALARY   |
+-------------+------------------+----------+
|         132 | TJ Olson         |  2100.00 |
|         128 | Steven Markle    |  2200.00 |
|         136 | Hazel Philtanker |  2200.00 |
|         127 | James Landry     |  2400.00 |
|         135 | Ki Gee           |  2400.00 |
|         119 | Karen Colmenares |  2500.00 |
|         131 | James Marlow     |  2500.00 |
|         140 | Joshua Patel     |  2500.00 |
|         168 | Lisa Ozer        | 11500.00 |
|         147 | Alberto Errazuriz | 12000.00 |
|         108 | Nancy Gruenberg  | 12008.00 |
|         205 | Shelley Higgins  | 12008.00 |
|         201 | Michael Martinez | 13000.00 |
|         146 | Karen Partners   | 13500.00 |
|         145 | John Singh       | 14000.00 |
|         101 | Neena Yang       | 17000.00 |
|         102 | Lex Garcia       | 17000.00 |
|         100 | Steven King      | 24000.00 |
+-------------+------------------+----------+
107 rows in set (0.001 sec)
```

4. Display first name and joining date of the employees who are either IT Programmer or Sales Man.

```
mysql> SELECT FIRST_NAME, HIRE_DATE, JOB_ID FROM employees WHERE JOB_ID IN ('IT_PROG', 'SA_MAN');
+------------+------------+---------+
| FIRST_NAME | HIRE_DATE  | JOB_ID  |
+------------+------------+---------+
| Alexander  | 2016-01-03 | IT_PROG |
| Bruce      | 2017-05-21 | IT_PROG |
| David      | 2015-06-25 | IT_PROG |
| Valli      | 2016-02-05 | IT_PROG |
| Diana      | 2017-02-07 | IT_PROG |
| John       | 2014-10-01 | SA_MAN  |
| Karen      | 2015-01-05 | SA_MAN  |
| Alberto    | 2015-03-10 | SA_MAN  |
| Gerald     | 2017-10-15 | SA_MAN  |
| Eleni      | 2018-01-29 | SA_MAN  |
+------------+------------+---------+
10 rows in set (0.001 sec)
```

5. Display details of employees with employee ID 150 or 160.

```
mysql> SELECT EMPLOYEE_ID, FIRST_NAME FROM employees WHERE EMPLOYEE_ID IN ('150', '160');
+-------------+------------+
| EMPLOYEE_ID | FIRST_NAME |
+-------------+------------+
|         150 | Sean       |
|         160 | Louise     |
+-------------+------------+
2 rows in set (0.001 sec)
```

6. Display first name, salary, commission pct, and hire date for employees with salary less than 10000.

```
mysql> SELECT FIRST_NAME, SALARY, HIRE_DATE, COMMISSION_PCT FROM employees WHERE SALARY < 10000;
+-------------+---------+------------+----------------+
| FIRST_NAME  | SALARY  | HIRE_DATE  | COMMISSION_PCT |
+-------------+---------+------------+----------------+
| Alexander   | 9000.00 | 2016-01-03 |           NULL |
| Bruce       | 6000.00 | 2017-05-21 |           NULL |
| David       | 4800.00 | 2015-06-25 |           NULL |
| Valli       | 4800.00 | 2016-02-05 |           NULL |
```

```
| Donald      | 2600.00 | 2017-06-21 |           NULL |
| Douglas     | 2600.00 | 2018-01-13 |           NULL |
| Jennifer    | 4400.00 | 2013-09-17 |           NULL |
| Pat         | 6000.00 | 2015-08-17 |           NULL |
| Susan       | 6500.00 | 2012-06-07 |           NULL |
| William     | 8300.00 | 2012-06-07 |           NULL |
+-------------+---------+------------+----------------+
88 rows in set (0.001 sec)
```

7. Display employees where the first name or last name starts with S.

```
mysql> SELECT FIRST_NAME, LAST_NAME FROM employees WHERE FIRST_NAME LIKE 'S%' OR LAST_NAME LIKE 'S%';
+------------+-----------+
| FIRST_NAME | LAST_NAME |
+------------+-----------+
| Sundar     | Ande      |
| Shelli     | Baida     |
| Sarah      | Bell      |
| Shelley    | Higgins   |
| Susan      | Jacobs    |
| Steven     | King      |
| Sundita    | Kumar     |
| Steven     | Markle    |
| Samuel     | McLeod    |
| Nandita    | Sarchand  |
| Ismael     | Sciarra   |
| John       | Seo       |
| Sarath     | Sewall    |
| John       | Singh     |
| Lindsey    | Smith     |
| William    | Smith     |
| Stephen    | Stiles    |
| Martha     | Sullivan  |
| Patrick    | Sully     |
| Sigal      | Tobias    |
| Sean       | Tucker    |
| Shanta     | Vollman   |
+------------+-----------+
22 rows in set (0.001 sec)
```

8. Display details of jobs in the descending order of the title.

```
mysql> SELECT JOB_ID FROM employees ORDER BY JOB_ID DESC;
+------------+
| JOB_ID     |
+------------+
| ST_MAN     |
| ST_MAN     |
| ST_MAN     |
| ST_MAN     |
| ST_MAN     |
| ST_CLERK   |
| ST_CLERK   |
```

```
| FI_ACCOUNT |
| FI_ACCOUNT |
| FI_ACCOUNT |
| AD_VP      |
| AD_VP      |
| AD_PRES    |
| AD_ASST    |
| AC_MGR     |
| AC_ACCOUNT |
+------------+
107 rows in set (0.001 sec)
```

9. Display the details of the employees from department 30, earning the salary greater 10000 but do not receive the commission.

```
mysql> SELECT DEPARTMENT_ID, FIRST_NAME, SALARY, COMMISSION_PCT FROM employees WHERE DEPARTMENT_ID = 30 AND SALARY > 10000 AND COMMISSION_PCT IS NULL;
```

```
mysql> SELECT FIRST_NAME, MANAGER_ID FROM employees WHERE MANAGER_ID IN (100, 120);
+------------+------------+
| FIRST_NAME | MANAGER_ID |
+------------+------------+
| Neena      |        100 |
| Lex        |        100 |
| Den        |        100 |
| Matthew    |        100 |
| Adam       |        100 |
| Payam      |        100 |
| Shanta     |        100 |
| Kevin      |        100 |
| John       |        100 |
| Karen      |        100 |
| Alberto    |        100 |
| Gerald     |        100 |
| Eleni      |        100 |
| Michael    |        100 |
| Julia      |        120 |
| Irene      |        120 |
| James      |        120 |
| Steven     |        120 |
| Winston    |        120 |
| Jean       |        120 |
| Martha     |        120 |
| Girard     |        120 |
+------------+------------+
22 rows in set (0.001 sec)
```

10. Display the details of employees reporting to the managers with employee id 100 and 120.

```
mysql> SELECT FIRST_NAME, MANAGER_ID FROM employees WHERE MANAGER_ID IN (100, 120);
+------------+------------+
| FIRST_NAME | MANAGER_ID |
+------------+------------+
| Neena      |        100 |
| Lex        |        100 |
| Den        |        100 |
| Matthew    |        100 |
| Adam       |        100 |
| Payam      |        100 |
| Shanta     |        100 |
| Kevin      |        100 |
| John       |        100 |
| Karen      |        100 |
| Alberto    |        100 |
| Gerald     |        100 |
| Eleni      |        100 |
| Michael    |        100 |
| Julia      |        120 |
| Irene      |        120 |
| James      |        120 |
| Steven     |        120 |
| Winston    |        120 |
| Jean       |        120 |
| Martha     |        120 |
| Girard     |        120 |
+------------+------------+
22 rows in set (0.001 sec)
```

11. Display the unique country_id from locations table.

```
mysql> SELECT DISTINCT COUNTRY_ID FROM locations;
+------------+
| COUNTRY_ID |
+------------+
| AU         |
| BR         |
| CA         |
| CH         |
| CN         |
| DE         |
| GB         |
| IN         |
| IT         |
| JP         |
| MX         |
| NL         |
| SG         |
| US         |
+------------+
14 rows in set (0.001 sec)
```

12. Display all employees whose have the job_id as IT_PROG and FI_ACCOUNT.

```
mysql> mysql> SELECT FIRST_NAME, JOB_ID FROM employees WHERE JOB_ID IN('IT_PROG', 'FI_ACCOUNT');
+-------------+------------+
| FIRST_NAME  | JOB_ID     |
+-------------+------------+
| Daniel      | FI_ACCOUNT |
| John        | FI_ACCOUNT |
| Ismael      | FI_ACCOUNT |
| Jose Manuel | FI_ACCOUNT |
| Luis        | FI_ACCOUNT |
| Alexander   | IT_PROG    |
| Bruce       | IT_PROG    |
| David       | IT_PROG    |
| Valli       | IT_PROG    |
| Diana       | IT_PROG    |
+-------------+------------+
10 rows in set (0.001 sec)
```

13. Display all countries in ascending order.

```
mysql> SELECT * FROM countries ORDER BY COUNTRY_NAME;
+------------+----------------------------------------+-----------+
| COUNTRY_ID | COUNTRY_NAME                           | REGION_ID |
+------------+----------------------------------------+-----------+
| AR         | Argentina                              |        20 |
| AU         | Australia                              |        40 |
| BE         | Belgium                                |        10 |
| BR         | Brazil                                 |        20 |
| CA         | Canada                                 |        20 |
| CN         | China                                  |        30 |
| DK         | Denmark                                |        10 |
| EG         | Egypt                                  |        50 |
| FR         | France                                 |        10 |
| DE         | Germany                                |        10 |
| IN         | India                                  |        30 |
| IL         | Israel                                 |        30 |
| IT         | Italy                                  |        10 |
| JP         | Japan                                  |        30 |
| KW         | Kuwait                                 |        30 |
| ML         | Malaysia                               |        30 |
| MX         | Mexico                                 |        20 |
| NL         | Netherlands                            |        10 |
| NG         | Nigeria                                |        50 |
| SG         | Singapore                              |        30 |
| CH         | Switzerland                            |        10 |
| GB         | United Kingdom of Great Britain and Nort |        10 |
| US         | United States of America               |        20 |
| ZM         | Zambia                                 |        50 |
| ZW         | Zimbabwe                               |        50 |
+------------+----------------------------------------+-----------+
25 rows in set (0.001 sec)
```